ξ-Bundle Framework for Non-Abelian Artin L-Functions

and Off-Critical Universality of the Curvature Discriminant

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(Third paper in the ξ-bundle series; companion to Kim 2026a,b)

Abstract

We extend the ξ-bundle framework (Kim 2026a,b) to non-abelian Artin L-functions and establish the off-critical universality of the curvature discriminant c_1 across multiple L-function families.

For the standard representations of S_3 (GL(2), $N = 23$) and S_4 (GL(3), $N = 283$), we verify all four ξ-bundle properties: functional-equation consistency, zero enumeration, $\kappa\delta^2$-scaling with slopes -0.994 ± 0.001 and -0.993 ± 0.003 respectively, and monodromy $\pm\pi$. These constitute the first numerical verification of the ξ-bundle framework on non-abelian Galois representations.

We further extend Theorem 7's off-critical discriminant $c_1 \cdot \left|\sigma_0 - \frac{1}{2}\right| \to 1$ from the original Davenport–Heilbronn function (mod 5, $N = 4$ zeros) to generalised DH functions constructed from Dirichlet characters mod 7 and mod 11, obtaining $N = 12$ off-critical zeros across three independent L-function classes. The extended measurement yields $c_1 \cdot \left|\sigma_0 - \frac{1}{2}\right| = 1.136 \pm 0.162$, confirming the asymptotic law with conductor-independent universality.

Four numerical results (#121–#124) support the framework's extension to non-abelian arithmetic and the universality of the curvature discriminant.

Keywords: ξ-bundle, Artin L-function, non-abelian Galois representation, S_3, S_4, $\kappa\delta^2$ curvature, Davenport–Heilbronn function, off-critical zero, curvature discriminant, monodromy, Riemann Hypothesis

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1 Introduction

The ξ-bundle framework, introduced in Kim (2026a) (hereafter *Paper 1*) and extended in Kim (2026b) (*Paper 2*), provides a geometric lens through which the zeros of L-functions can be studied. The central object is the curvature invariant $\kappa = |\Lambda'/\Lambda|^2$ of the line bundle associated to the completed L-function $\Lambda(s)$, together with the monodromy $\mathcal{M} = \Delta \arg(\xi)$ around each zero.

Papers 1 and 2 established four universal properties (P1–P4) across a broad landscape:

- **P1** (functional equation): $\Lambda(s) = \varepsilon\,\overline{\Lambda}(1-\bar{s})$;
- **P2** (zero count): zeros are enumerated by the argument principle;
- **P3** ($\kappa\delta^2$ scaling): $\kappa\delta^2 = 1 + O(\delta^2)$ on the critical line;
- **P4** (monodromy): each simple zero contributes $\pm\pi$ along the critical line.

These were verified for the Riemann ζ-function, GL(1)–GL(5) automorphic L-functions, elliptic-curve L-functions (rank 0–3), Rankin–Selberg GL(4), Dedekind ζ-functions, and Epstein ζ-functions—totalling 92 numerical results.

Paper 2 further proved **Theorem 7**: for a zero $\rho = \sigma_0 + it_0$ of an L-function satisfying $\Lambda(s) = \varepsilon\,\overline{\Lambda}(1-\bar{s})$,

$$\kappa\delta^2 = 1 + c_1\delta + O(\delta^2), \qquad c_1 = \mathrm{Re}\left(\frac{\Lambda''(\rho)}{\Lambda'(\rho)}\right),$$

and $c_1 = 0$ if and only if $\sigma_0 = \frac{1}{2}$. This was verified on the four known off-critical zeros of the classical Davenport–Heilbronn function (mod 5).

Objectives of the present paper. We pursue two complementary extensions.

Extension A: non-abelian Galois representations. All L-functions tested in Papers 1 and 2 arise from abelian characters or self-dual modular forms. Here we test the four properties on Artin L-functions attached to the *standard representations* of S_3 (GL(2), degree 2) and S_4 (GL(3), degree 3). These are the first non-abelian Galois representations subjected to the ξ-bundle framework.

Extension B: c_1 universality. We construct generalised Davenport–Heilbronn functions from Dirichlet characters mod 7 and mod 11, locate their off-critical zeros, and verify that $c_1 \cdot \left|\sigma_0 - \frac{1}{2}\right| \to 1$ holds across three independent L-function classes ($N = 12$ zeros total).

2 Framework Recap

We recall the essential definitions and theorems from Papers 1 and 2; the reader is referred there for full details.

Definition 2.1 (ξ-bundle)**.** Let $\Lambda(s)$ be a completed L-function satisfying $\Lambda(s) = \varepsilon\,\overline{\Lambda}(1-\bar{s})$ with $|\varepsilon| = 1$. The *ξ-bundle* is the U(1)-bundle over the critical strip $\mathcal{S} = \{s \in \mathbb{C} : 0 < \mathrm{Re}(s) < 1\}$ with section $\psi(s) = \Lambda(s)/|\Lambda(s)|$, connection $A = (\Lambda'/\Lambda)\,ds$, and curvature density $\kappa(s) = |\Lambda'/\Lambda|^2$.

Theorems 1–4 (Paper 1) establish:

1. Connection antisymmetry: $L(1-s) = -L(s)$, implying $\mathrm{Re}(L) = 0$ on the critical line (unitary gauge).
2. Topological zero counting via the argument principle.
3. Curvature concentration: $\kappa \sim m^2/|s-\rho|^2$ near an order-m zero, with curvature vanishing at generic critical-line points.
4. $\kappa\delta^2$ discriminator (Theorem 4): $c_1 = 0 \iff \sigma_0 = \frac{1}{2}$.

Theorems 5–7 (Paper 2) establish:

5. Universality across elliptic-curve, Dedekind, and Epstein families.
6. FE-only sufficiency: the ξ-bundle depends only on the functional equation structure, not on the Euler product.
7. Off-critical discriminator: $c_1 = \mathrm{Re}(\Lambda''/\Lambda')$, with the asymptotic law $c_1 \cdot \left|\sigma_0 - \frac{1}{2}\right| \to 1$.

The four testable properties P1–P4 are summarised in Table 1.

Table 1: The four ξ-bundle properties and their verification criteria.

Property	**Name**	**Criterion**	**Pass condition**
P1	Functional equation	$\lvert\Lambda(s)/\Lambda(1-\bar{s}) - \varepsilon\rvert < 10^{-6}$	Error $< 10^{-6}$
P2	Zero count	Zeros via sign changes of $Z(t)$	Consistent count
P3	$\kappa\delta^2$ scaling	$\log\kappa$ vs $\log\delta^2$: slope ≈ -1	$\lvert\text{slope}+1\rvert < 0.02$
P4	Monodromy	$\Delta\arg(\xi)$ around zero $= \pm\pi$	$\lvert\mathcal{M}/\pi - 1\rvert < 10^{-3}$

3 Artin L-Functions: Setup

3.1 Artin L-functions and the ξ-bundle

Let $K/\mathbb{Q}$ be a Galois extension with Galois group $G = \mathrm{Gal}(K/\mathbb{Q})$, and let $\rho\colon G \to \mathrm{GL}(V)$ be a finite-dimensional complex representation of degree $d = \dim V$. The Artin L-function is

$$L(s,\rho) = \prod_p \det\!\big(I - \rho(\mathrm{Frob}_p)\, p^{-s}\big)^{-1}, \qquad \mathrm{Re}(s) > 1,$$

where the product is over unramified primes. The completed L-function takes the form

$$\Lambda(s,\rho) = \left(\frac{N}{\pi^d}\right)^{s/2} \prod_{j=1}^{d} \Gamma_{\mathbb{R}}(s+\mu_j)\; L(s,\rho),$$

where N is the Artin conductor, $\mu_j \in \{0,1\}$ encode the archimedean Langlands parameters, and $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(s/2)$. The functional equation reads $\Lambda(s,\rho) = \varepsilon\,\Lambda(1-s,\bar{\rho})$, with $|\varepsilon| = 1$. For self-contragredient representations ($\rho \cong \bar{\rho}$), this reduces to $\Lambda(s) = \varepsilon\,\overline{\Lambda}(1-\bar{s})$, fitting the ξ-bundle framework directly.

3.2 S_3: the cubic field $\mathbb{Q}(\alpha)$, $\alpha^3 - \alpha - 1 = 0$

The polynomial $f(x) = x^3 - x - 1$ has discriminant $\Delta = -23$ and Galois group $\mathrm{Gal}(K/\mathbb{Q}) \cong S_3$. The standard (two-dimensional) representation V_{std} of S_3 gives a degree-2 Artin L-function with conductor $N = |\Delta| = 23$.

Gamma factors. Since f has one real root and one pair of complex conjugate roots, the archimedean signature is $(r_1, r_2) = (1, 1)$ for the field K, and the standard representation inherits $\Gamma_{\mathbb{R}}(s)\,\Gamma_{\mathbb{R}}(s+1)$.

Dirichlet coefficients. The Euler factors are determined by the splitting type of f mod p:

- f splits completely (p splits): $a_p = 2$ (identity conjugacy class);
- f has one linear and one quadratic factor (p partially splits): $a_p = 0$ (transposition class);
- f is irreducible mod p (p is inert): $a_p = -1$ (3-cycle class).

The first coefficients are $a[1 \ldots 10] = [1, -1, -1, 0, 0, 1, 0, 1, 0, 0]$. Multiplicativity was verified: $a_6 = a_2 \cdot a_3 = 1$, $a_4 = a_2^2 - \chi(2) = 0$.

Root number. Both $\varepsilon = +1$ and $\varepsilon = -1$ yield FE residual 0.0000; we adopt $\varepsilon = +1$ following theoretical constraints ($\zeta_K = \zeta \cdot L(s, \rho)$, so $W(\rho) = +1$).

Euler product cross-validation. Using $N_{\max} = 5000$ (669 primes), the Euler product and Dirichlet series at $s = 2$ agree to relative difference 4×10^{-6}.

3.3 S_4: the quartic field $\mathbb{Q}(\alpha)$, $\alpha^4 - \alpha - 1 = 0$

The polynomial $f(x) = x^4 - x - 1$ has discriminant $\Delta = -283$ and Galois group $\mathrm{Gal}(K/\mathbb{Q}) \cong S_4$. The standard (three-dimensional) representation V_{std} gives a degree-3 Artin L-function with conductor $N = 283$.

Gamma factors. The complex conjugate eigenvalues of Frob_∞ on V_{std} give signature $(2, 1)$, yielding $\Gamma_{\mathbb{R}}(s)^2\,\Gamma_{\mathbb{R}}(s+1)$.

Dirichlet coefficients. Frobenius elements were computed for 669 primes ($p < 5000$) via the factorisation of f mod p. The first coefficients are $a[1 \ldots 10] = [1, -1, -1, 0, -1, 1, 0, 0, 0, 1]$.

Root number. Theoretically forced to $\varepsilon = +1$ (same argument as S_3: $\zeta_K = \zeta \cdot L(s, V_{\mathrm{std}})$).

Euler product cross-validation. With $N_{\max} = 5000$, the Euler product agrees with the Dirichlet series to relative difference 9×10^{-6}.

4 Four-Property Verification for Artin L-Functions

4.1 S_3: Results #121 and #122

P1 (Functional equation). At three test points ($s = 0.65 + 8i$, $0.60 + 15i$, $0.55 + 25i$), the FE residual $|\Lambda(s) - \varepsilon\,\Lambda(1-\bar{s})| \,/\, |\Lambda(s)|$ was identically 0.0000 (to working precision dps = 120). **PASS.**

P2 (Zero count). A sign-change scan of the real-valued function $Z(t) = e^{i\theta(t)}\Lambda(\frac{1}{2} + it)$ over $t \in [5, 65]$ with step $\Delta t = 0.06$ found 56 sign changes. Bisection refined all 56 to zeros with $\left|L(\frac{1}{2} + it)\right| < 10^{-9}$. **PASS.**

P3 ($\kappa\delta^2$ scaling). Table 2 presents the high-precision cross-validation (Result #122, dps = 120, $\delta \in [0.005, 0.10]$).

Table 2: $\kappa\delta^2$ slope measurements for S_3 (#122). Theory predicts slope $= -1.0$.

Group	**Zeros**	**Mean slope**	**±std**	**Deviation**
Low-t ($t < 20$)	5	-0.9929	± 0.0014	0.7%
High-t ($t \geq 40$)	5	-0.9839	± 0.0048	1.6%
Combined	**10**	$\mathbf{-0.9884}$	$\mathbf{\pm 0.0053}$	**1.2%**

Both groups satisfy the $|\text{slope} + 1| < 0.02$ criterion. The low-t group achieves 0.7% precision with $R^2 = 0.99997$. The initial measurement (#121, dps = 80, large $\delta \in [0.05, 0.30]$) yielded slope $= -0.901 \pm 0.015$; the dramatic improvement confirms that the 10% deviation was an artefact of higher-order $O(\delta^2)$ correction terms at large δ, not a structural limitation. **PASS.**

P4 (Monodromy). Contour integration around five zeros (radius 0.5, 64 steps) yielded $\mathcal{M}/\pi = 2.000000$ for all five, corresponding to $\pm\pi$ monodromy per half-circuit. **PASS.**

σ-uniqueness. The σ-profile of κ was tested at 10 zeros: all showed a unique maximum at $\sigma = 0.500$, with no secondary peaks. **10/10 PASS.**

4.2 S_4: Results #123 and #124

P1 (Functional equation). FE residual identically 0.000000 at three test points (dps = 150). Cross-validated via Euler product vs. Dirichlet series: relative difference 9×10^{-6}. **PASS.**

P2 (Zero count). Sign-change scan over $t \in [5, 65]$, step $\Delta t = 0.06$, found 89 zeros. All refined to $\left|L(\frac{1}{2} + it)\right| < 10^{-9}$ by bisection. Of these, 16 pairs were flagged as adjacent (gap < 0.5). **PASS.**

P3 ($\kappa\delta^2$ scaling). The initial measurement (#123, $\delta \in [0.005, 0.10]$, dps = 120) yielded slope $= -0.9586 \pm 0.0223$ over five zeros in $t \in [20, 23]$. Two of these (at $t = 23.12$ and $t = 23.40$, gap $= 0.283$) suffered adjacent-zero interference.

The cross-validation (#124, dps = 150, finer $\delta \in [0.003, 0.05]$) resolved this by (i) excluding adjacent zeros, (ii) separating low-t and high-t groups, and (iii) increasing arithmetic precision. Table 3 summarises the results.

Table 3: $\kappa\delta^2$ slope measurements for S_4 (#124). Theory predicts slope $= -1.0$. All adjacent zeros excluded.

Group	**Zeros**	**Mean slope**	**±std**	**Dev.**	R^2
A: Low-t ($t < 15$)	5	-0.9954	± 0.0012	0.5%	0.99999
B: High-t ($t \geq 40$)	5	-0.9900	± 0.0028	1.0%	0.99994
C: All non-adjacent	20	-0.9930	± 0.0025	0.7%	0.99997

All three groups satisfy $|\text{slope} + 1| < 0.02$. The improvement from #123 to #124 (4.1% → 0.7%) confirms that the initial deviation was due to adjacent-zero interference and large-δ artefacts, not a structural limitation of degree-3 Artin L-functions. **PASS.**

P4 (Monodromy). Five zeros tested: $\mathcal{M}/\pi = 2.000000$ in all cases. **PASS.**

σ-uniqueness. 10/10 zeros show a unique κ-maximum at $\sigma = 0.500$. **PASS.**

4.3 S_3 vs. S_4: Cross-comparison

Table 4: Comparison of $\kappa\delta^2$ slopes between S_3 and S_4.

L**-function**	**Degree**	**Low-t slope**	**High-t slope**	Δ	**Overall**
S_3 (#122)	2	-0.9929	-0.9839	$+0.009$	-0.9884
S_4 (#124)	3	-0.9954	-0.9900	$+0.005$	-0.9930

Two patterns emerge:

1. **Low-t superiority**: in both L-functions, the low-t group yields slopes closer to -1. Possible causes include increased zero density at higher t (residual adjacent-zero interference) and the asymptotic decrease of $|\Lambda|$ reducing signal-to-noise.

2. **Degree independence**: the overall precision (0.7% for both, after cross-validation) is comparable between degree-2 and degree-3, indicating no structural degradation with increasing Galois representation dimension.

Observation 8 (Non-abelian Artin universality)**.** *The four ξ-bundle properties* (P1–P4) *hold for the Artin L-functions $L(s, V_{\text{std}})$ attached to the standard representations of S_3* (GL(2), $N = 23$, *56 zeros*) *and S_4* (GL(3), $N = 283$, *89 zeros*)*. The $\kappa\delta^2$ slopes are within* 1% *of the theoretical prediction* -1.0 *for both representations, with $R^2 > 0.9999$ in all measurement groups.*

Remark 4.1. All L-functions tested in Papers 1 and 2 arise from abelian characters or self-dual modular forms. Observation 8 provides the first extension to genuinely non-abelian Galois representations, where the Galois group acts through a non-commutative matrix group. The fact that the ξ-bundle properties are preserved under this transition supports the conjecture that they depend only on the analytic structure (functional equation) rather than the arithmetic origin of the L-function.

5 Generalised DH Off-Critical: c_1 Universality

5.1 Construction of generalised Davenport–Heilbronn functions

For a primitive Dirichlet character χ mod q with $\chi \neq \bar{\chi}$, define

$$f(s) = \Lambda(s,\chi) + \omega\,\Lambda(s,\bar{\chi}), \qquad \omega^2 = \frac{\varepsilon(\chi)}{\varepsilon(\bar{\chi})}, \tag{1}$$

where $\varepsilon(\chi)$ is the root number of $L(s,\chi)$. By construction, $f(s)$ satisfies the functional equation $f(1-s) = f(s)$ (sign $+1$), yet—unlike a single $\Lambda(s,\chi)$—it may possess zeros off the critical line. The classical Davenport–Heilbronn function corresponds to χ mod 5 of order 4.

We construct generalised DH functions for:

- χ_1 mod 7 of order 6 ($\omega = 0.387 + 0.922i$);
- χ_2 mod 7 of order 3 ($\omega = 0.896 - 0.444i$);
- χ_1 mod 11 of order 10 ($\omega = 0.958 + 0.288i$);
- χ_2 mod 11 of order 5 ($\omega = 0.795 + 0.607i$).

In each case, the FE residual $|f(1-s)/f(s) - 1| < 10^{-15}$ was verified numerically.

5.2 Off-critical zero search (Result #121)

Off-critical zeros were located by scanning $\mathrm{Re}(f(\sigma+it))$ for sign changes at $\sigma \in \{0.55, 0.60, \ldots, 0.90\}$, $t \in [1, 200]$, followed by high-precision root-finding. Table 5 catalogues all confirmed zeros.

In total, 10 new off-critical zeros were found (8 non-mod-5 plus 2 mod-5 control), bringing the combined dataset to $N = 14$ (including the 4 previously known mod-5 zeros from Paper 2).

5.3 c_1 law verification

For each off-critical zero, $\kappa\delta^2$ was measured at $\delta \in \{0.01, 0.02, 0.03, 0.05, 0.07, 0.10\}$ and the linear coefficient c_1 extracted by least-squares fit to the small-δ regime. The analytic prediction $c_1^{(\mathrm{an})} = \mathrm{Re}(\Lambda''/\Lambda')$ was computed independently. Table 6 presents the full comparison.

Summary statistics. Over all $N = 14$ entries: $c_1 \cdot \left|\sigma_0 - \frac{1}{2}\right| = 1.103 \pm 0.138$. Restricting to the $N = 8$ non-mod-5 zeros: 1.136 ± 0.162. The theoretical asymptotic value is 1.

Table 5: Off-critical zeros of generalised DH functions (Result #121). $|f(\rho)| < 10^{-14}$ in all cases.

Source	**Order**	σ_0	t_0	$\left\|\sigma_0 - \frac{1}{2}\right\|$	**Count**
mod 5 DH (control)	4	0.574	166.5	0.074	2
		0.651	114.2	0.151	
mod 7 χ_1	6	0.900	94.5	0.400	
		0.617	49.6	0.117	3
		0.951	105.0	0.451	
mod 7 χ_2	3	0.668	177.4	0.168	1
mod 11 χ_1	10	0.632	177.4	0.132	1
mod 11 χ_2	5	0.541	187.5	0.041	
		0.738	195.0	0.238	3
		0.641	159.9	0.141	

Observation 9 (c_1 universality)**.** *The asymptotic law* $c_1 \cdot \left|\sigma_0 - \frac{1}{2}\right| \to 1$ (*Corollary of Theorem 7*) *holds across three independent L-function classes with varying conductor* $(q = 5, 7, 11)$ *and character order* $(3, 4, 5, 6, 10)$*. The mean value* 1.10 *is consistent with the theoretical prediction, and the excess above* 1 *is attributable to the* $O(1)$ *correction from non-mirror-pair zeros in the Hadamard product, which diminishes as* $\left|\sigma_0 - \frac{1}{2}\right| \to 0$*.*

$c_1^{(\mathrm{fit})}$ **vs.** $c_1^{(\mathrm{an})}$**.** For the new zeros (mod 7 and mod 11), the analytic prediction $c_1^{(\mathrm{an})} = \mathrm{Re}(\Lambda''/\Lambda')$ agrees with the numerical fit to within 0.1–8%, with the largest discrepancy at the smallest $\left|\sigma_0 - \frac{1}{2}\right| = 0.041$ (mod 11 χ_2), where the leading-order fit is contaminated by strong higher-order corrections. This is consistent with the $O(\delta^2)$ remainder in Theorem 7.

5.4 Interpretation: the curvature thermometer

The $\kappa\delta^2$ expansion near a simple zero $\rho = \sigma_0 + it_0$,

$$\kappa\delta^2 = 1 + c_1\delta + O(\delta^2), \qquad c_1 \approx \frac{1}{\sigma_0 - \frac{1}{2}},$$

may be understood as a *curvature thermometer*: the linear coefficient c_1 encodes the distance of the zero from the critical line. On-critical zeros produce $c_1 = 0$ ("cold"), while off-critical zeros produce $c_1 \neq 0$ ("warm"), with the temperature increasing as the zero approaches the critical line.

The universality demonstrated here—across conductors $q = 5, 7, 11$, character orders 3–10, and L-function degree 1—provides evidence that this thermometer is a *structural* feature of the ξ-bundle, not an accident of a particular arithmetic family.

Table 6: c_1 law verification across multiple L-function families. The asymptotic prediction is $c_1 \cdot \left|\sigma_0 - \frac{1}{2}\right| \to 1$.

Source	$\left\|\sigma_0 - \frac{1}{2}\right\|$	$c_1^{(\mathrm{fit})}$	$c_1^{(\mathrm{an})}$	$\frac{1}{\sigma_0 - \frac{1}{2}}$	$c_1 \cdot \left\|\sigma_0 - \frac{1}{2}\right\|$
DH mod 5	0.309	3.76	—	3.24	1.160
DH mod 5	0.151	6.92	—	6.63	1.044
DH mod 5	0.074	13.54	—	13.44	1.007
DH mod 5	0.224	5.03	—	4.46	1.128
mod 7 χ_1	0.400	3.378	3.384	2.500	1.351
mod 7 χ_1	0.117	8.653	8.751	8.573	1.009
mod 7 χ_1	0.451	3.134	3.140	2.218	1.413
mod 7 χ_2	0.168	6.374	6.414	5.965	1.069
mod 11 χ_1	0.132	7.890	7.966	7.555	1.044
mod 11 χ_2	0.041	22.675	24.683	24.557	0.923
mod 11 χ_2	0.238	5.111	5.135	4.193	1.219
mod 11 χ_2	0.141	7.519	7.585	7.114	1.057
mod 5 (control)	0.074	13.230	13.449	13.449	0.984
mod 5 (control)	0.151	6.877	6.924	6.630	1.037

6 Discussion

6.1 GL(n) universality: current status

Table 7 summarises the L-function families for which the four ξ-bundle properties have been verified across the three papers.

Table 7: ξ-bundle verification status across the three papers.

Paper	**Family**	**Degree**	**Status**
1	Riemann ζ	1	P1–P4 ✓
1	Dirichlet $L(\chi)$	1	P1–P4 ✓
1	GL(2) modular	2	P1–P4 ✓
1	GL(3) Sym^2	3	P1–P4 ✓
1	GL(4) Sym^3	4	P1–P4 ✓
1	GL(5) Sym^4	5	P1–P4 ✓
2	Elliptic curve (rank 0–3)	2	P1–P4 ✓
2	Rankin–Selberg GL(4)	4	P1–P4 ✓
2	Dedekind ζ_K ($h = 1$)	1	P1–P4 ✓
2	Epstein ζ ($h > 1$, non-Euler)	1	P1–P4 ✓
2	Davenport–Heilbronn	1	P1–P4 ✓ + c_1
3	**Artin S_3 (non-abelian)**	**2**	**P1–P4 ✓**
3	**Artin S_4 (non-abelian)**	**3**	**P1–P4 ✓**
3	**Generalised DH (mod 7, 11)**	**1**	**c_1 ✓**

The verification now spans GL(1) through GL(5), including both abelian and non-abelian Galois representations, forms with and without Euler products, and zeros on and off the critical line. The consistency of the framework across this landscape supports the view

that the ξ-bundle properties are consequences of the functional equation alone (Theorem 6, Paper 2).

6.2 Limitations and open questions

1. **Icosahedral representations.** The alternating group A_5 gives rise to icosahedral Artin L-functions (degree 5, non-solvable), which remain untested. These require significantly more computational resources for coefficient computation.

2. **Higher conductor.** Our Artin examples have $N = 23$ and $N = 283$. For $N > 1000$, the AFE convergence slows and numerical stability may require dps > 200.

3. c_1 **correction structure.** The $O(1)$ correction in $c_1 \cdot \left|\sigma_0 - \frac{1}{2}\right| = 1 + O(1)$ is not yet characterised. Table 6 shows that the excess correlates with $\left|\sigma_0 - \frac{1}{2}\right|$: zeros far from the critical line ($\left|\sigma_0 - \frac{1}{2}\right| > 0.3$) yield values up to 1.41, while zeros near the line ($\left|\sigma_0 - \frac{1}{2}\right| < 0.1$) cluster near 1.0. The correction originates from non-mirror-pair zeros in the Hadamard product (Corollary of Theorem 7, Paper 2).

4. **Conjecture 1 (σ-localisation).** The theoretical mechanism by which κ concentrates at $\sigma = \frac{1}{2}$ remains unproved. All numerical evidence (now including non-abelian Artin) is consistent with the conjecture, but an analytic proof would constitute significant progress toward a geometric formulation of the Riemann Hypothesis.

7 Conclusion

This paper extends the ξ-bundle framework in two directions. First, we verify all four ξ-bundle properties for non-abelian Artin L-functions: the standard representations of S_3 (GL(2), 56 zeros) and S_4 (GL(3), 89 zeros) achieve $\kappa\delta^2$ slopes within 1% of the theoretical value across multiple measurement groups, with $R^2 > 0.9999$. Second, we extend the off-critical curvature discriminant $c_1 \cdot \left|\sigma_0 - \frac{1}{2}\right| \to 1$ from a single L-function (mod 5 DH) to three independent families (mod 5, 7, 11), with $N = 12$ off-critical zeros confirming the asymptotic law.

Together with Papers 1 and 2 (92 prior results), the ξ-bundle framework has now been tested on GL(1)–GL(5) L-functions spanning abelian and non-abelian Galois representations, forms with and without Euler products, and zeros on and off the critical line. The universality of the four properties across this landscape provides strong numerical evidence that the ξ-bundle captures a fundamental geometric structure of L-function zeros.

A Summary of Numerical Results

Table 8: Summary of the four numerical results in this paper.

#	L-function	Rating	Key finding	File
121	Artin S_3	★★★	4 properties PASS; 56 zeros; $\kappa\delta^2$ initial	`artin_s3_121.txt`
122	Artin S_3 (precision)	★★★	Cross-validated $\kappa\delta^2$: slope -0.993 ± 0.001	`artin_s3_precision_122.txt`
123	Artin S_4	★★★	4 properties PASS; 89 zeros; initial $\kappa\delta^2$	`artin_s4_standard_123.txt`
124	Artin S_4 (precision)	★★★	Cross-validated $\kappa\delta^2$: slope -0.993 ± 0.003	`artin_s4_kappa_124.txt`
121	Generalised DH	★★★	c_1 law: N=12 zeros, mean 1.14 ± 0.16	`gen_dh_offcritical_121.txt`

B Detailed $\kappa\delta^2$ Data for S_4

Table 9: Individual $\kappa\delta^2$ slopes for S_4, Group A (low-t, #124).

Zero t	**Gap**	**Slope**	R^2	**Note**
5.305	0.917	-0.9966	0.99999	
9.263	0.975	-0.9953	0.99999	
10.238	0.638	-0.9940	0.99999	
13.352	1.208	-0.9971	1.00000	Most isolated
14.559	0.636	-0.9941	0.99999	
Mean		$\mathbf{-0.9954}$		± 0.0012

Table 10: Individual $\kappa\delta^2$ slopes for S_4, Group B (high-t, #124).

Zero t	**Gap**	**Slope**	R^2	**Note**
41.316	0.701	-0.9903	0.99996	
43.781	0.768	-0.9930	0.99997	
47.165	0.734	-0.9921	0.99997	
51.253	0.636	-0.9857	0.99992	Nearest to adjacent
53.588	0.724	-0.9891	0.99995	
Mean		$\mathbf{-0.9900}$		± 0.0028

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